



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

CONCENTRATION VACUUM PRODUCING EQUIPMENT (X-2301)

Docu. No. : 14780-8120-PS-X2301
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

CONCENTRATION VACUUM PRODUCING EQUIPMENT

(X-2301)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-X2301**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

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PROCESS SPECIFICATION



PTT PHENOL TRAIN II PROJECT
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Record of Revision

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VACUUM PUMPS AND DRIVERS

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	4JUN12	MW	DRL				
1	5JUN13	DRL	MEH				

UOP Phenol Process Unit Concentration, Decomposition and Neutralization Section

Item Number	Service	Sheets
X-2301	Concentration Vacuum Producing Equipment	3 - 13
	Appendix – Equipment Sketch	14
	Attachment E-2310 : Flash Column Overhead Vapor Chiller Datasheet (For reference)	15

Supplier (or equal): Gardner Denver Nash

www.gdnash.com

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 7-12

UOP Standard Drawing:

UOP Project Specification: 963766-401

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	4JUN12	MW	DRL				
1	5JUN13	DRL	MEH				

Ejector Design Conditions*

I. Concentration Vacuum Producing Equipment to be Furnished

Furnish a hybrid vacuum producing equipment package consisting of:

1. Deleted.
2. Item No. X-2301 Service: Concentration Vacuum Producing Equipment
 - a. Single stage steam jet ejector* (1x100% and 2x50%) (1 set required)
 - b. Chilled Water cooled intercondenser (1 required)
 - c. Single stage Liquid Ring Vacuum Pump (LRVP) with full recirculation sealant liquid system (2x100%)

* The ejector manufacturer shall also provide a set of standard auxiliaries, including a steam strainer, steam purifier/filter and trap, steam separator, and interconnecting piping. Steam and ejector gate valves will be furnished by the ejector manufacturer. A steam letdown valve shall be provided (0.7 kg/cm² minimum pressure drop). "Go/No-Go" gauges shall be provided on the steam lines for ejector protection.

The POSCO ENGINEERING is responsible for all interconnecting piping outside of piping provided on skid by LRVP manufacturer. POSCO ENGINEERING to determine scope break.

II. By Contrator

Deleted.

III. Process Controls

The process control is automatic control of Flash and Preflash Columns overhead pressure from a 4-20 mA signal from the pressure controller (PIC-1002) to a spillback control valve (PV-1002). Spillback take-off is from the ejector discharge upstream of intercondenser, and recycling back to the ejector inlet. (See sketch on sheet 14)

The normal operating pressure of the Flash Column Overhead Vapor Chiller (Item No. E-2310) is 11.5 mm Hg (a). Refer to the P&ID 14780-8120-25-23-0010 and Process Flow Diagram (PFD) 14780-8120-20-23-0003 of this project.

The LRVP systems will have dedicated spillback control valves on recycle lines from the discharge separator to the LRVP suction for suction pressure control. Spillbacks shall be sized for 100% machine capacity. Vendor shall determine the requirements for the spillback and size equipment accordingly

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0	4JUN12	MW	DRL				
1	5JUN13	DRL	MEH				

Ejector Design Conditions*

The design loads to the ejectors indicated below are 110% of the process requirement.

UOP proposes two parallel ejector elements be provided. The first element shall be sized for 100% of the Design load indicated below. The second element shall be sized for 50% of the Design Load indicated below.

Isolation valves shall be provided on the load gas suction, motive steam supply and ejector discharge for each ejector element to allow for inspection of steam nozzle and diffuser throat during operation.

The ejector elements shall be arranged such that the ejector elements (100%/50%) may be operated independently or simultaneously using a common intercondenser.

Ejectors shall be stable from zero load up to and including the design point.

	<u>Design</u>
Inlet Pressure, mm Hg (a)	10
Inlet Temperature, °C)	5
Ejector Outlet Pressure, mm Hg (a)	914
Vapor Source:	Preflash and Flash Columns Condenser

Non-condensables:

Nitrogen (MW = 28.01), kg/hr	57.2
Oxygen (MW = 32.0), kg/hr	<u>6.71</u>
Sub-Total, kg/hr	63.91

Condensables:

Formic Acid (MW = 46.03), kg/hr	7.59
Water (MW = 18), kg/hr	66.88
Cumene (MW = 120), kg/hr	99.44
AMS (MW = 118.18), kg/hr	0.009
Acetone (MW = 58), kg/hr	<u>0.77</u>
Sub-Total, kg/hr	174.69

Total, kg/hr **238.6**

* - Vacuum system supplier and EPC Contractor shall update vapor load data to the ejectors based on the vapor outlet load from the Flash Column Overhead Vapor Chiller.

* - Refer to attached datasheet of Flash Column Overhead Vapor Chiller.



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VACUUM PUMPS AND DRIVERS

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	4JUN12	MW	DRL				
1	15AUG12	MW	MEH				
2	5JUN13	DRL	MEH				

Estimated System Volume, m³ 700

Approximate System Start-up evacuation time: 4 hours. Vendor to advise in proposal actual time required to evacuate system.

Vapor Line to Ejectors, inches Vapor Discharge Line of
Flash Column Overhead Chiller : 18"
Size of Vapor Line to Ejectors : 12" and 10"
Precondenser Outlet Temperature 10°C max.

Intercondenser condensate liquid shall be drained to the Decanter, Item No D-2205. The rundown pipe to the Decanter is sealed by 14000 mm minimum of liquid (reference note in Section II. Sheet 4.). Provide an additional intercondenser shell vent for high capacity non-condensable removal during startup (2").

Intercondenser shall be designed for 150% load when both the 50% and the 100% ejector elements are in operation.

Utilities

<u>Steam at S12 (MP) Header</u>	<u>Minimum</u>	<u>Normal</u>	<u>Maximum</u>	<u>Design</u>
Pressure, kg/cm ² (g)	11.5	12	12.5	21
Temperature, °C	189	191	205	250

Steam at Ejector Inlet

Pressure, kg/cm ² (g)	10.8 (0.7kg/cm ² across letdown valve)
Temperature, °C (minimum)	191

Estimated Steam and Water Consumptions:

Ejector Steam	570 kg/hr (For 150% operation, 855 kg/hr)
Intercondenser Chilled Water	93.9 m ³ /hr

<u>Chilled Water</u>	<u>Supply</u>	<u>Return</u>	<u>Design</u>
Pressure at grade, kg/cm ² (g)	6.75	2.5	11.5
Temperature, °C	0	5	120
MDMT, °C	Reference Section XI. Sheet 8		

Ejector Elevation EL + 33.84 m from ground

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0	4JUN12	MW	DRL				
1	5JUN13	DRL	MEH				

V. Ejector Materials of Construction

Ejector Motive Nozzle	Stainless Steel, Type 304L
Ejector Steam Chest	304L SS
Suction Chamber and Diffusers	304L SS

VI. Design and Construction of Intercondensers

Materials of Construction

Tubes and Tubesheet (Stationary and Floating)	304L SS
Channel and Channel Cover	KCS (Std. 3-11)
Baffles, Tube Supports, Tie Rods, and Spacers	304 SS
Shell and Shell Cover	304L SS (Std. 3-15)
Floating Head Cover/Bolts	304L SS (Std. 3-15)

TEMA Type and Class

AET, Class R

Tubes

Seamless: 19.05 mm O.D.,
Min.wall thickness 1.65 mm

Tube Length

3770 mm

Tube Pitch

25.4 mm

Allowable Pressure Drop – Cooling Water

0.7 kg/cm²

Allowable Pressure Drop – Process Side

0.006 kg/cm²

Minimum Fouling Factor

Cooling Water Side	0.0004 h-m ² -°C/kcal
Steam Side	0.0003 h-m ² -°C/kcal

Notes:

- The intercondenser of this hybrid vacuum system shall be designed and constructed in accordance with UOP Specification 4-11. It shall be free draining from the shell side inlet to outlet.
- Deleted.

Dip-pipes and Interconnecting Piping

Dip leg and interconnecting piping shall be 304L stainless steel. The contractor shall confirm that the dip-pipe velocity does not exceed 0.5 m/s.



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1	5JUN13	DRL	MEH				

VII. Painting

The supplier shall provide equipment with one coat of paint on the outside surface before leaving the shop. Inlets, outlets, and other connections shall be sealed and protected with weather-proof shipping paper to prevent rusting of internal surfaces.

VIII. Design Pressures

Ejector:

Internal: 21 kg/cm² (g) and 250°C

External: Full Vacuum

Intercondensers:

Process Side: Internal: 4 kg/cm² (g) and 250°C

External: Full Vacuum

Water Side: Internal: 11.5 kg/cm² (g) and 120°C

MDMT = Reference Section XI below.

IX. Strainers and Valves

Steam strainers and steam valves shall have steel bodies. Bronze or cast iron bodies are not acceptable.

X. Construction to Standards

Construction shall be in accordance with the latest edition of Standards for Steam Jet Ejectors and Standard for Steam Surface Condensers - Heat Exchange Institute, unless otherwise specified herein (removable tube bundle type).

XI. Minimum Design Metal Temperature

The minimum design metal temperature for all equipment is to use the lower value of either 15 °C or 14 °C below the operating temperature.

XII. GENERAL INFORMATION

Elevation of plant 23.0 m
Normal Barometer 760 mm Hg
Ambient Temperature Range 15°C to 38°C



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VACUUM PUMPS AND DRIVERS

REV	DATE	BY	APVD	REV	DATE	BY	APVD
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1	5JUN13	DRL	MEH				

Final Stage Mechanical Vacuum Producing Equipment

Type: Single Stage Liquid Ring Vacuum Pump (LRVP)

No. of Pumps Required: 2 x 100% capacity **

Drivers: Fixed Speed Motor: Direct Connected

Fluid Pumped: See Sheet 11

DESIGN*

Normal m³/hr at 760 mm Hg (a) and 0°C *

Mass Flow at Suction Conditions, kg/hr *

Flow at Suction Conditions, m³/hr 790

Suction Temperature, °C 8.3

Suction Pressure, mm Hg (a) 79.75

Molecular Weight *

Final Discharge Pressure, mm Hg (a) 804

Final Discharge Temperature, °C 18

Materials and General Comments See Sheets 10 through 12

Est. RPM 980 rpm (Vendor to confirm)

Est. kW Required @ Design Capacity, kW 29 kW (Vendor to confirm)

Motor Comments See Sheet 13

Recommended Motor Size, kW The motors shall be sized for a minimum of 115% of the rated power requirements for the LRVP.

LRVP Recirculating Sealant Flow, m³/hr 3.43

Est. Sealant liquid purge rate, m³/hr 10-15 lb/hr Hydrocarbons (Vendor to confirm)

10-15 lb/hr Water (Vendor to confirm)

* - The Vacuum producing equipment vendor shall establish the gas rates based on the non-condensable gas leaving the intercondenser between the ejectors and the LRVP pump for this hybrid solution.

** - The LRVP shall be sized for 110% of the governing case.

The LRVP shall have no negative tolerance on performance.

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
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1	5JUN13	DRL	MEH				

Description of 2nd Stage Mechanical Vacuum Producing Equipment:

Furnish a mechanical vacuum compression package for the final stage of this hybrid vacuum system. The final stage shall consist of 2x100% single stage Liquid Ring Vacuum Pumps (LRVP) and drivers with full recirculation sealant liquid system. Each LRVP shall be sized for 110% of the design process requirements.

As a minimum, the package shall include a discharge separator (sized to allow for phase separation and vapor-liquid disengagement), 2x100% 316 SS sealant liquid circulation pumps, 2x100% chilled water cooled sealant liquid coolers (Isolation valves and other accessories on cooling water by contractor), duplex circulating sealant liquid strainers, LRVP suction check valve, isolation valves per sheet 14 sketch, spillback valve for suction pressure control of the LRVP, 2x100% discharge pressure relief valves, provision for flow control of the sealant liquid makeup and separator level control for purge, isolation valve for vent gas dilution nitrogen line, low point drain connections for piping to the Aqueous Waste closed drain header including a valve on the drains of the LRVP, and a local operation panel.

Discharge pressure of LRVP liquid purge is approximately 4.12 kg/cm² including pumping losses to the Decanter. Contractor to supply actual pump requirements based on final system layout.

The system will operate in an outdoor, unheated area. The LRVP shall have dual mechanical seals and flanged suction and discharge connections. The suction and discharge flanges shall be Cl. 150 RF. The bearings shall be grease lubricated. The LRVP shall be mounted on a common base-plate with the sealant liquid system. All pressure vessels shall be designed in accordance with Section VIII, Division 1 of the ASME Pressure Vessel Code, but do not require a code stamp. The chilled water cooled sealant liquid cooler shall be TEMA class C with no removable bundle. A direct connected LRVP is preferred.

The LRVP shall include the following System Shutdowns using transmitters and separated from control instruments:

- High differential pressure alarm and shutdown.
- Low recirculated sealant liquid flow alarm and shutdown.
- High temperature alarm and shutdown on the inlet to the LRVP with a setting of 93°C (200°F) (for machine protection).
- High and low liquid level alarm and shutdown on discharge separator.

The contractor shall coordinate with the LRVP supplier the specifics of the LRVP shutdown signals.

The system shall be switchover from the main to the standby LRVP on manual, and permissive to start shall be initiated by local hand switch at field by field operator.

Spillback systems to limit the stage compression ratio and provide for a safe and stable start-up shall be included. Spillbacks shall be sized for 100% of machine capacity. Spillback valves shall be “fail closed” design. The contractor shall be responsible for coordinating details and defining the vendor scope of supply.

Vendor shall clearly state all requirements to air free the system before startup and hydrocarbon free the system after shutdown. Vendor shall provide additional connections as needed to meet these requirements.



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1	16AUG12	MW	MEH				
2	5JUN13	DRL	MEH				

LRVP Sealant Liquid

The sealant liquid makeup shall be process water from the CHP Process Water Pumps at the Oxidation section. Sealant purge shall be returned to the Decanter, where the normal operating pressure is .084 kg/cm² (g). Elevation of the Decanter is 7700mm. X-2301 will be located on the structure at 30300 mm from grade. Sealant circulation pump head to be determine by pump vendor considering requirement head for recirculation.

Sealant Liquid Composition (Weight Percent)

Water 100
Total 100

Flowing Density: 988 kg/m³

Molecular Weight: 18.

Supply Pressure at Pump: 6.76 kg/cm² (g)

Supply Temperature: 50°C

Est. Makeup Rate = 0.45 m³/hr

Expected Recirculated Sealant Liquid Composition (Weight Percent)

Water 45
Cumene 55
Total 100

Flowing Density: Vendor to advise. kg/m³

Molecular Weight: Vendor to advise.

Inlet Temperature: Vendor to advise. °C

Circulating flow = 3.43 m³/hr

Est. Purge Rate = 0.113 m³/hr

Vacuum Pump Design Details

CASINGS
ROTORS
BEARINGS
SHAFTS
PORT PLATE / CONNECTION PLATE
SHAFT SEALS
MDMT, °C

316 Stainless Steel
316 Stainless Steel
Anti-Friction with min L-10 life of 25,000 hrs
316 Stainless Steel
316 Stainless Steel / 316 Stainless Steel
Unpressurized Dual, mechanical type
Reference Section XI. Sheet 8

Discharge Separators

Provide a discharge separator with 2x 100% relief valves. The vessel shall be sized to ensure satisfactory phase separation of any condensable as well as vapor/liquid disengagement and include coalescing type mist elimination on the vent. The vessel shall have a minimum of a 2 minute surge time between bottom of inlet nozzle and the maximum liquid level. The vendor shall state the actual gas velocity in the separator vapor space. The vendor shall state the maximum liquid carryover, and vessel diameter in his proposal. The vessel shall include a 1" nitrogen nozzle for vent gas dilution. High and low liquid level alarm and shutdown transmitters shall be included. The vessel shall be made of 304L stainless steel and designed and constructed in accordance with ASME Code, Section VIII, Division 1.

A local level gauge and level controller shall be included.

The design temperature shall be 120°C and the design pressure shall be 3.5 kg/cm² (g). Reference section XI, sheet 8 for MDMT. The minimum corrosion allowance is 1.5 mm.

Reference UOP Std. Specification 3-15: Pressure Vessels – Austenitic Steel

The vessel shall be designed to ensure phase separation of increasing vapor/liquid load during parallel operation for LRVP switchover and Start-up.

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
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1	5JUN13	DRL	MEH				

Circulating Chilled Water – See sheet 6

Nitrogen for Dilution

CHP Nitrogen will be used for vent gas dilution in the discharge separators to maintain an oxygen concentration less than 5% (lower flammability limit). Nitrogen lines to the discharge separator will include isolation valves. Nitrogen design conditions are shown below.

		CHP N ₂
Minimum Operating Pressure:	kg/cm ² (g)	
Normal Operating Pressure:	kg/cm ² (g)	7
Maximum Operating Pressure:	kg/cm ² (g)	9
Mechanical Design Pressure:	kg/cm ² (g)	10

Accessories

All accessories and special tools required for proper installation, operation, and maintenance shall be furnished by the vendor.

Instrumentation

Instrument Air Pressure	7 kg/cm ² (g) Normal
Instrument Power	220V / Single phase / 50HZ
Instrument Graduating System	MKS, Pressure in kg/cm ² (g) and mm HG

Vendor shall provide instrumentation as shown in the sketch on sheet 14, as a minimum. See legend on P&ID 963763-120-01-A1 for instrumentation symbol identification.

Baseplate

Provide a common baseplate for the pumps and drivers and sealant recirculation system. Baseplates shall include drip lips and drains.

Motor Starter and Pushbutton Station

Provided by others.

Serviceman

Vendor shall quote the cost per 8-hour day for a competent serviceman to supervise the installation, loading of sealing liquid, and placing equipment in operation. Purchaser shall pay all traveling and living expenses.

Spare Parts

The vendor shall provide a list of recommended spare parts for commissioning and two years operation.

Testing

The vendor shall conduct a shop performance test on each unit using all package components, if practical. The capacity shall have a zero negative tolerance and the power tolerance is +4%. Certified test data for the pumps shall be made available to UOP and the contractor.

Relief Valve

The vendor shall furnish 2x100% relief valves located on the discharge separator at the outlet of the LRV. Valve shall be located below the mesh blanket, with the outlet connected to the Decanter at Oxidation Section.



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1	5JUN13	DRL	MEH				

Discharge Isolation Valve

The POSCO ENGINEERING shall provide an automatic discharge isolation valve. This valve shall be fail close. It will receive a signal from vacuum equipment shutdown system and close whenever the LRVP is de-energized.

The POSCO ENGINEERING shall review the details of this system and coordinate implementation.

Electric Motors

Elevation above sea level: 23.0 m

<u>kW RANGE</u>	<u>Voltage</u>	<u>Phase</u>	<u>Hertz</u>
0 to 159 kW	380	3	50
160 kW and Up	6600	3	50

Enclosure Types: Ex 'n' IIA T3, IP55

Refer to 14780-38A1 : Squirrel Cage Induction Motors & IEC 60034

Note: Motors shall be suitable for a coastal and salty sea air environment.

Instrumentation

Instrument Air Pressure	7 kg/cm ² (g) Normal (Min. 4.0 kg/cm ² g / Max. 7.5 kg/cm ² g)
Instrument Power	24V DC
Instrument Graduating System	MKS, Pressure in kg/cm ² (g) and mm HG

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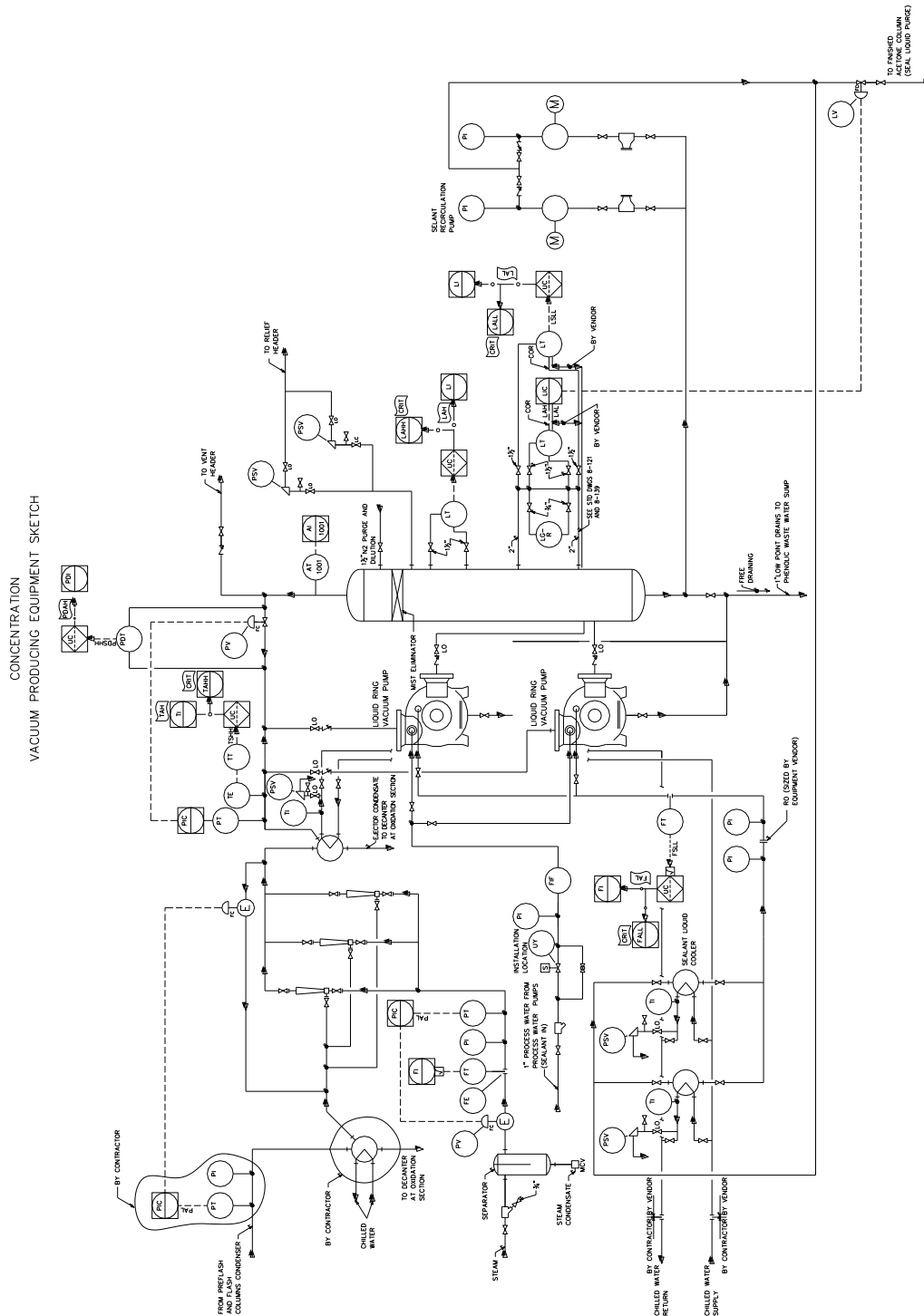
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Appendix – Equipment Sketch



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SHEET 15 / 15

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Attachment: Flash Column Overhead Vapor Chiller Datasheet(For reference)

				14780-8480-DS-0010 REV D1 Page 2 of 9 Item Number E-2310			
SHELL AND TUBE EXCHANGER THERMAL DATA SHEET System of Units Metric							
Case	DESIGN	TEMA Type	AEU	Orientation	Horizontal	Rev	
Number of Units	1	Shells per Unit	1	Shells in Series	1	D1	
Surface per Unit m²	224.4	Surface per Shell m²	224.4	Shells in Parallel	1	D1	
PERFORMANCE OF ONE UNIT				In SHELLSIDE Out		In TUBESIDE Out	
Fluid Name	Hydrocarbon			Chilled Water (Note 4)			
Total Flowrate (Note 2)	kg/hr	3085 X 1.1		64472 X 1.1		D1	
Vapour	kg/hr	3027 X 1.1	231.4 X 1.1	0	0	D1	
Liquid	kg/hr	0	2795.6 X 1.1	64472 X 1.1	64472 X 1.1	D1	
Fluid Condensed	kg/hr	2795.6 X 1.1		0		D1	
Fluid Evaporated	kg/hr	0		0		D1	
Noncondensibles	kg/hr	58 x 1.1				D1	
LIQUID PROPERTIES				SEE PAGES 6 AND 7		SEE PAGE 8	
Density	kg/m³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Surface Tension	dyne/cm						
Pour Point Temperature	°C						
VAPOUR PROPERTIES				SEE PAGES 6 AND 7			
Density	kg/m³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Hydrate Temperature	°C						
THERMAL AND HYDRAULIC DATA							
Operating Temperature	°C	38.0	5.0	0.1	5.0	D1	
Operating Pressure	kg/cm²g	-1.0163		6.750			
Number of Passes per Shell		1		8			
Average Velocity	m/s	16.8		1.0		D1	
Pressure Drop (Allow / Calc)	kg/cm²	0.00141 (Note 1)	0.00169 (Note 1)	0.703 (Note 20)	0.714	D1	
Fouling Factor	h.m².°C/kcal	0.0003		0.0004			
Duty (Note 2)	kcal/h X 10 ⁶	0.2927 X 1.1		MTD °C	18.6	D1	
U-value Service/Clean	kcal/h.m².°C	82	88	Over Surface %	6.5	D1	
DESIGN CONDITIONS				SHELLSIDE		TUBESIDE	
Design Pressure max/min	kg/cm²g	4.5		11.5		D1	
Full Vacuum		Required	at 38 °C	Not Required			
Test Pressure	kg/cm²g	To Code		To Code			
Design Temperature max/min	°C	120	-9	120	-14	D1	
Corrosion Allowance	mm	1.5		3	(for KCS)		
Insulation Details		Cold Conservation		Cold Conservation			
Hydrogen Service		No		No			
Sour Service		No		No			
Lethal Service		No		No			
Cyclic Service		No		No			
Cleaning Method		Mechanical		Mechanical			
Stress Relief for Process Reasons		No		No			
Heat Release Curve		Yes		Yes			